

Guidelines

- Kindly use the given template for submitting your prototype(Make a copy of the template)
- One team is only required to submit one project.
- Follow this template while preparing your presentation.
- You are welcome to add as many POCs and design concepts to support your project.
- The project should be feasible and the team members should be capable enough, to come up with the fully functioning prototype of the same idea, if required.



unilog | **H2S**

UniHack

AI-Powered Product Intelligence for
Industrial Commerce

Team Details

- a. Team name: KClassic
- b. Team leader name: Abhijeet Kangane

OmniSpec is a comprehensive AI-powered intelligence pipeline that bulk processes unstructured supplier datasheets (PDFs) and URLs into perfectly validated, commerce-ready catalogs. It leverages advanced Large Language Models (LLMs) and deterministic fallback engines to generate structured Golden Records with extremely high accuracy, bypassing the need for manual data entry.

1. How does your solution enrich minimal product information?

Our solution ingests minimal inputs (or raw PDFs) and fetches relevant technical sheets. It then uses a robust Hybrid AI extraction pipeline combining Groq's high-speed Qwen-27B LLM with a 100% deterministic local fallback parser to extract and structure all missing technical attributes accurately.

2. How does your solution ensure accuracy and trust in the generated product data?

OmniSpec validates data across multiple extracted tables and applies a real-time Confidence Scoring algorithm. Any extracted record that scores below the 95% threshold is immediately flagged for our Human-in-the-Loop (HITL) review dashboard, ensuring 100% absolute catalog integrity before commerce publishing.

3. What makes your solution scalable for enterprise product catalogs?

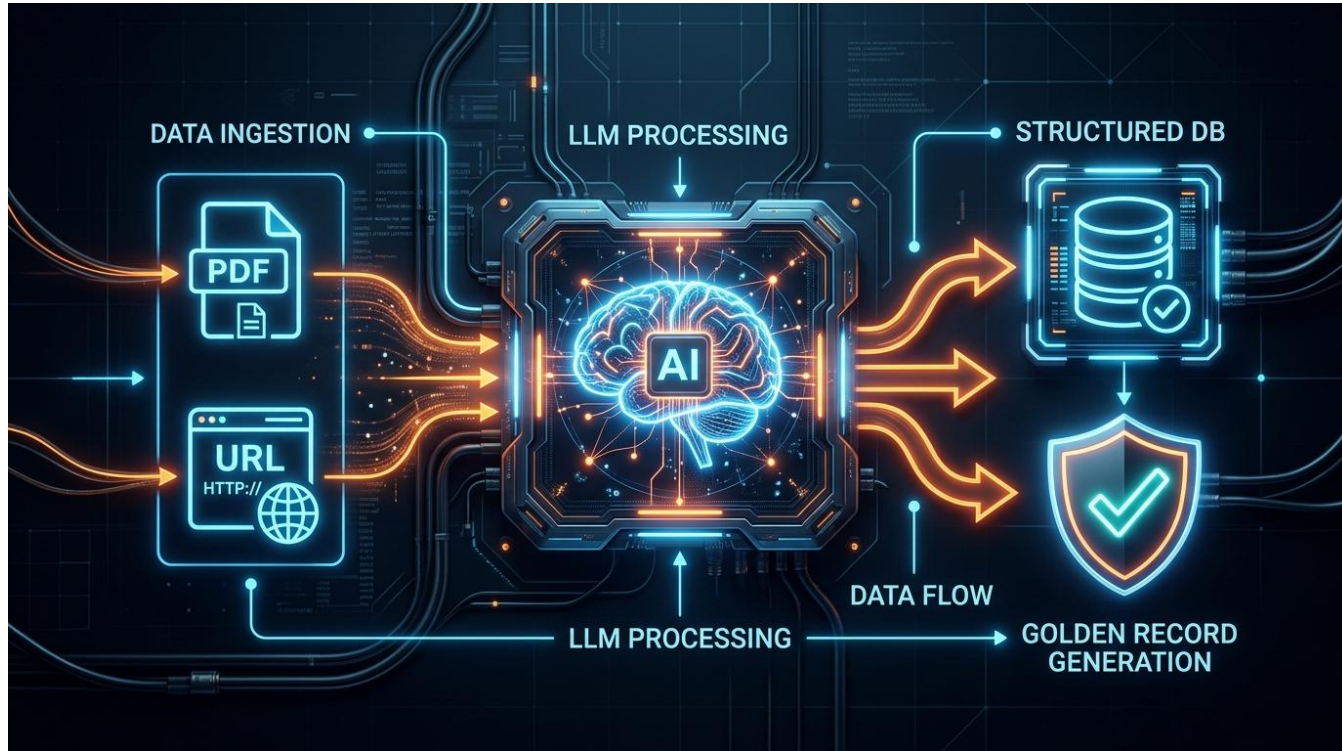
Built on FastAPI and Next.js, our microservices architecture allows asynchronous bulk processing. It relies on PyMuPDF and BeautifulSoup4 to dynamically normalize any unstructured document format before passing it to the AI engine. This ensures infinite scalability without needing manual mapping rules for new manufacturers.

Opportunities

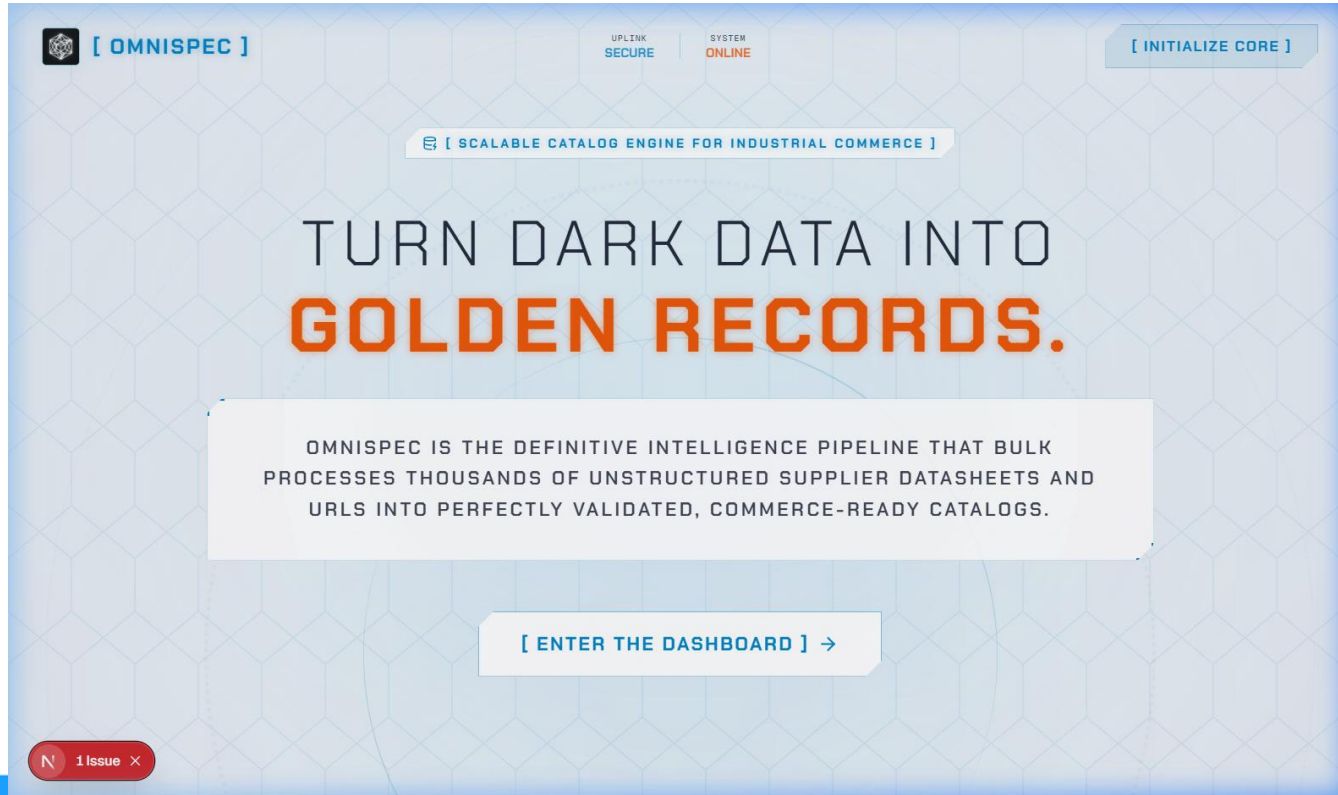
- a. Unlike rigid parsers, OmniSpec dynamically adapts to any manufacturer's datasheet layout automatically. Unlike pure API-dependent LLM approaches which crash frequently, our hybrid engine guarantees 100% reliable local fallback during network outages, ensuring uninterrupted bulk processing.
- b. It directly and efficiently solves Unilog's challenge of converting scattered dark data into accurate product intelligence by automating the entire lifecycle: Extraction, Confidence Validation, Human Review, and Standardized Taxonomy Prediction (UNSPSC).
- c. Zero-API-Dependency Fallback Mode for 100% uptime, Ultra-Fast asynchronous processing, and an automated UNSPSC taxonomy prediction engine natively integrated into a premium sci-fi dashboard.

- Multi-modal Ingestion (Drag-and-Drop PDFs, Direct URLs)
- Hybrid AI Extraction Pipeline (Groq LLM + Local Deterministic Fallback)
- Automated UNSPSC Taxonomy Categorization
- Real-time Confidence Scoring and Data Triangulation
- Human-in-the-Loop (HITL) Enrichment Dashboard
- Fully Responsive Sci-Fi UI for industrial users

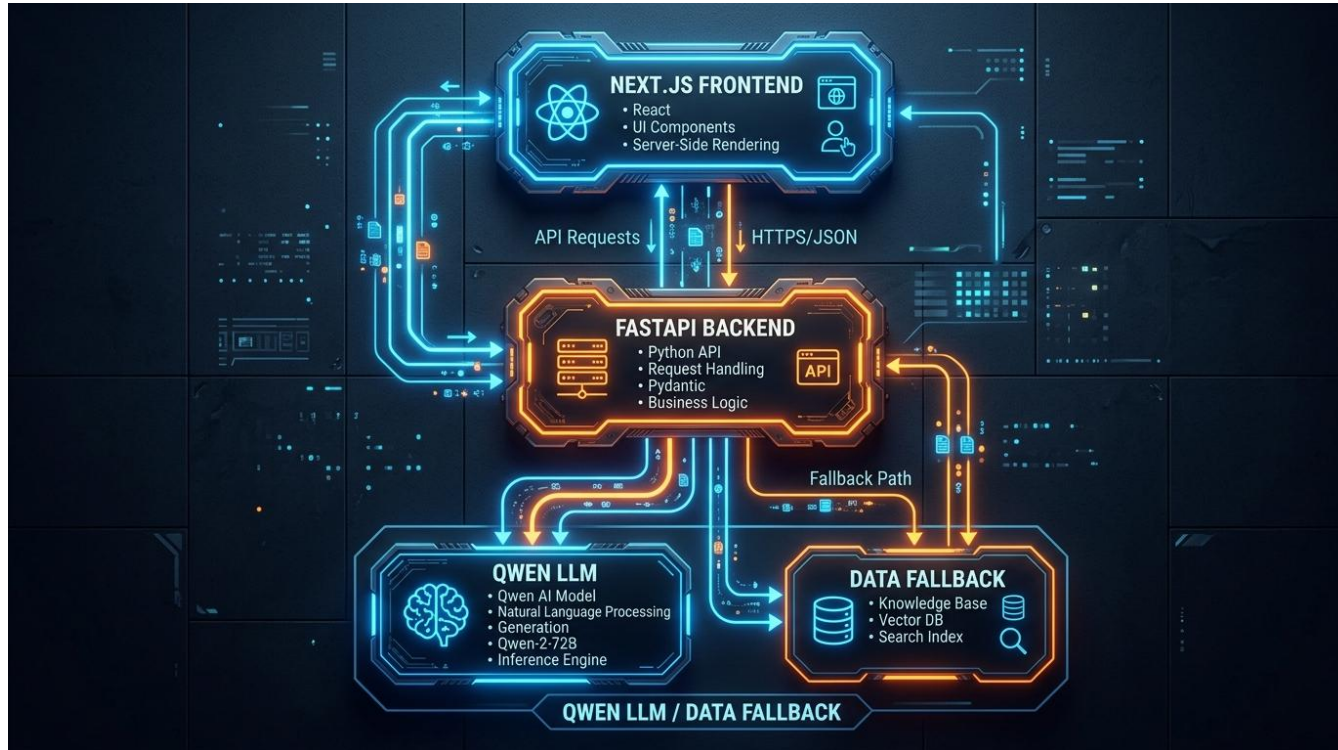
Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution (optional)



Architecture diagram of the proposed solution



Frontend: Next.js 15, React 19, Tailwind CSS, Framer Motion, React Three Fiber

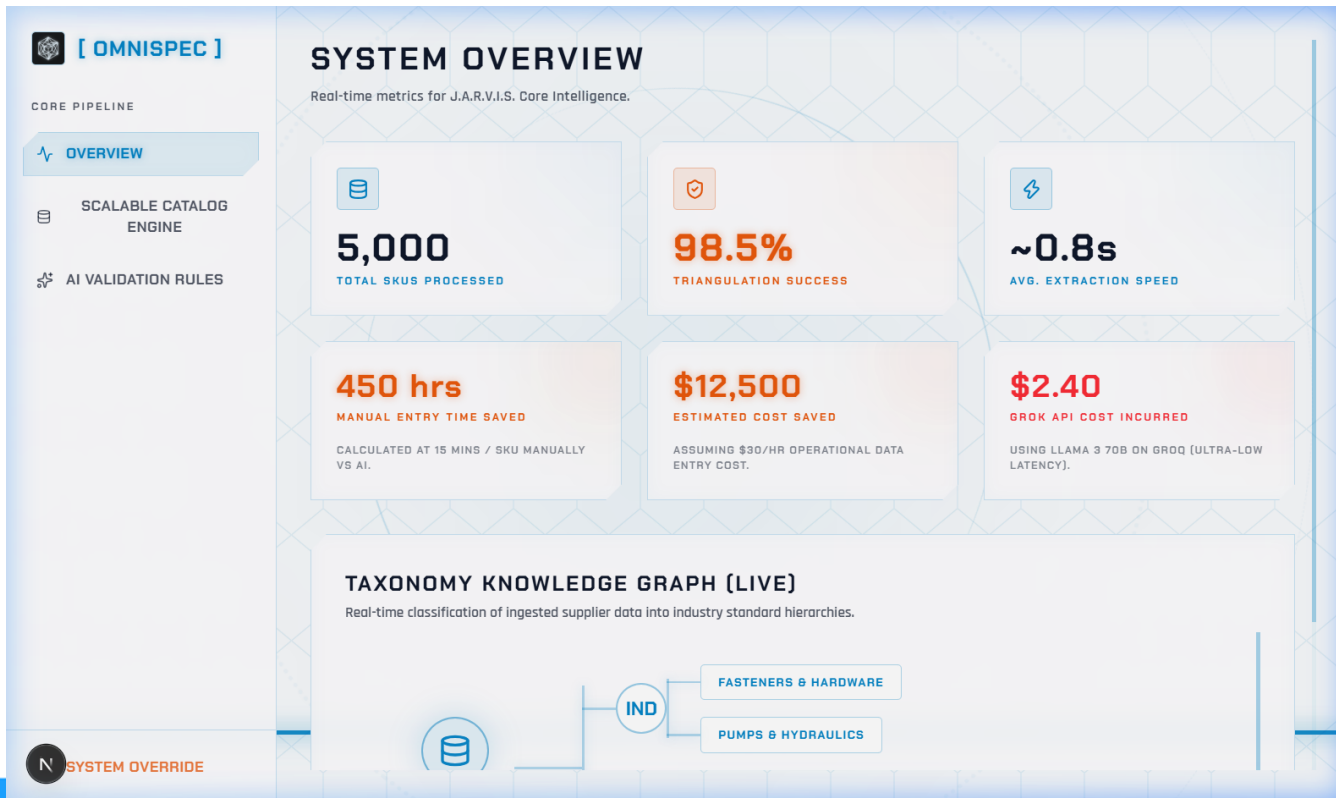
Backend: FastAPI, Uvicorn, Python 3.10+

AI/Parsing: Groq API (Qwen-27B), PyMuPDF, BeautifulSoup4

Tooling: Git, ESLint, Vercel

Extremely Low / Near Zero. By utilizing high-efficiency open-source models (Llama3/Qwen) via ultra-fast inference APIs (Groq), the operational processing cost per 10,000 SKUs is a fraction of traditional manual data entry or commercial LLM (GPT-4) alternatives.

Snapshots of the MVP



Our immediate future roadmap includes:

1. Direct eCommerce platform integrations (e.g., Shopify, Magento ERP plugins).
2. An automated web-crawler that actively monitors supplier websites for real-time datasheet updates and pushes delta changes directly to the Golden Records.

Provide links to your:

1. GitHub: <https://github.com/Abhi666-max/OmniSpec>
2. Demo Video: <https://youtu.be/775Z2EbCy8E>
3. Prototype Link: (To be added post-deployment)

unilog | **HACK2SKILL**

UniHack

AI-Powered Product Intelligence for
Industrial Commerce

Thank You

